PH-214 Modern Physics Notes

Useful Math

Taylor series expansions:

$$\sin x = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$\tan x = \sum_{n=1}^{\infty} \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!} x^{2n-1} = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \cdots$$

$$(1+x)^m = \sum_{n=0}^{\infty} \binom{m}{n} x^n = 1 + mx + \frac{m(m-1)}{2}x^2 + \frac{m(m-1)(m-2)}{6}x^3 + \cdots$$

<u>Note:</u> For small x, higher order terms reduce to zero

Euler's Identity:
Use complex exponentials to manipulate complicated trig functions and complex amplitudes.

$$e^{ix} = \cos x + i \sin x$$

Visualizing propagating plane waves:
A 1D plane wave propagating in the $+z$ direction can be mathematically represented as:

$$\psi = f(kz - \omega t)$$

Where:

- $(kz - \omega t)$ [rad]:
  The argument represents the **phase** of the wave. A change in phase simply moves the wave forward or backward in space and time, without changing its shape, there are a few ways to see this which we will look at later.

- k [rad/m]:
  is the **wave number**, representing the spatial frequency of the wave, or how *spread out* the wave is in space. Since k scales the spatial variable z, a larger k means the wave advances through the same phase over a smaller change in z.

- ω [rad/s]:
  is the **angular frequency**, representing the temporal frequency of the wave, or how *spread out* the wave is in time. As in rotational kinematics, ω scales time t, and a larger ω means the wave advances through the same phase over a smaller change in t.

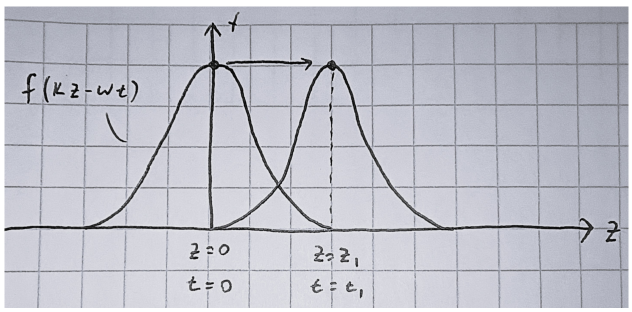
A plane wave is special in that it represents a wave whose shape is fixed as it propagates through space and time, with it's shape represented by the function f.

Note:

$$\boxed{k = \frac{2\pi}{\lambda}, \quad \frac{\omega}{2\pi} = f = \frac{1}{T}\,[\text{Hz}], \quad v = \frac{\omega}{k}}$$

Global View

As mentioned before, there are two ways to visualize how the phase $(kz - \omega t)$ represents a propagating wave. The first is a *global* view, looking at how z and t change relative to each other for a fixed phase, tracking a specific point on the wave.



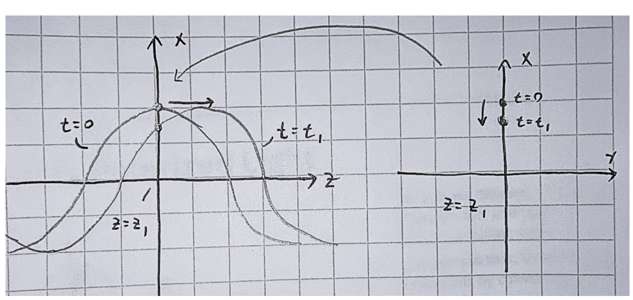
When we track a specific point on the wave, f must give the same value at different positions and times, meaning the phase must stay constant. This shows that as t increases, z must also increase to maintain the phase (the wave moves rightward!). Vice versa, as z increases, t must increase too to keep the phase constant (again, moving rightward!).

Note: This intuition applies to *any* point on the wave! Thus, by tracking all points, we can clearly see a propagating plane wave.

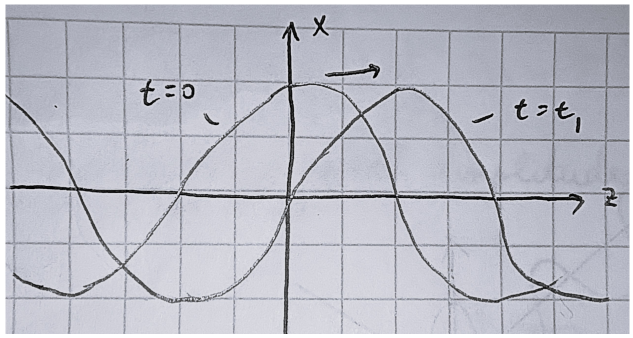
Local View

The second is a *local* view, looking at how the phase changes at a fixed position as time changes, or at a fixed time as position changes. For this, we will look at a sinusoidal wave, to better show oscillation: $\psi = cos(kz - \omega t)$

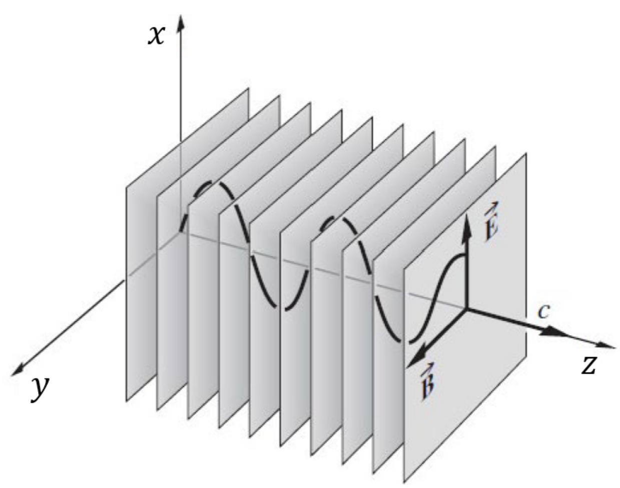
1. Fixing z, letting t vary:
   Imagine we are at a fixed position $z = z_1$. As time increases, we are effectively watching a cross-section of the wave at that point, seeing only a single point along the wave moving up and down like riding the "surface" of the wave.



2. Fixing t, letting z vary:
   Now imagine we are at a fixed time $t = t_1$. Moving along the z axis, and plotting the wave's amplitude, we see the full shape of the wave at that instant (ωt_1 becomes a constant phase shift).



3. Letting z and t vary together:
   When we looked at z and t separately, we saw only a *part* of the wave's behavior. For a fixed z, we observed how a single point moves over time, capturing the wave's temporal variation. For a fixed t, we saw the spatial variation, revealing the full shape of the wave statically at that instant. When both z and t vary together, we can see the wave's complete dynamic behavior, watching how its shape propagates through space and time.



Note: This intuition will become very useful when we study electromagnetic polarization, where the plane wave is a 2D wave. In our cross-sectional view, the single point along the wave will move along the xy plane instead of just up and down.

Vector Calculus

- **Del Operator:**
$$\vec{\nabla} = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$$

- **Laplacian Operator:**
$$\nabla^2 = \vec{\nabla} \cdot \vec{\nabla} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \quad \text{(in cartesian)}$$

- **Divergence:** Dot product of del operator with a vector field $\vec{A}$, giving a scalar field that represents how much $\vec{A}$ spreads (diverges) out from a point.
$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$
Note: A positive divergence indicates a source, while a negative divergence indicates a sink.

- **Curl:** Cross product of del operator with a vector field $\vec{A}$, giving a vector field that represents how much $\vec{A}$ curls around a point.
$$\begin{aligned}\vec{\nabla} \times \vec{A} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \\ &= \underbrace{\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)}_{\text{curl about } x\text{-axis}}\hat{x} + \underbrace{\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)}_{\text{curl about } y\text{-axis}}\hat{y} + \underbrace{\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)}_{\text{curl about } z\text{-axis}}\hat{z}\end{aligned}$$
Note: A positive curl indicates a counterclockwise rotation, while a negative curl indicates a clockwise rotation about the respective axis.

- **Vector Identities:**
Triple Products
$$\begin{aligned}\vec{A} \cdot (\vec{B} \times \vec{C}) &= \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B}) \\ \vec{A} \times (\vec{B} \times \vec{C}) &= \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})\end{aligned}$$
Product Rules
$$\begin{aligned}\vec{\nabla}(fg) &= f(\vec{\nabla}g) + g(\vec{\nabla}f) \\ \vec{\nabla}(\vec{A} \cdot \vec{B}) &= \vec{A} \times (\vec{\nabla} \times \vec{B}) + \vec{B} \times (\vec{\nabla} \times \vec{A}) + (\vec{A} \cdot \vec{\nabla})\vec{B} + (\vec{B} \cdot \vec{\nabla})\vec{A} \\ \vec{\nabla} \cdot (f\vec{A}) &= f(\vec{\nabla} \cdot \vec{A}) + \vec{A} \cdot (\vec{\nabla}f) \\ \vec{\nabla} \cdot (\vec{A} \times \vec{B}) &= \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B}) \\ \vec{\nabla} \times (f\vec{A}) &= f(\vec{\nabla} \times \vec{A}) - \vec{A} \times (\vec{\nabla}f) \\ \vec{\nabla} \times (\vec{A} \times \vec{B}) &= (\vec{B} \cdot \vec{\nabla})\vec{A} - (\vec{A} \cdot \vec{\nabla})\vec{B} + \vec{A}(\vec{\nabla} \cdot \vec{B}) - \vec{B}(\vec{\nabla} \cdot \vec{A})\end{aligned}$$

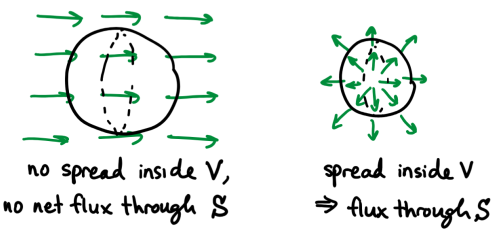
Second Derivatives

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0 \quad \text{(Divergence of a curl is always zero)}$$
$$\vec{\nabla} \times (\vec{\nabla} f) = 0 \quad \text{(Curl of a gradient is always zero)}$$
$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} \quad \text{(Curl of a curl)}$$

Divergence Theorem

Relates the flux of a vector field $\vec{F}$ through a closed surface S to the divergence of $\vec{F}$ within the volume V enclosed by S.

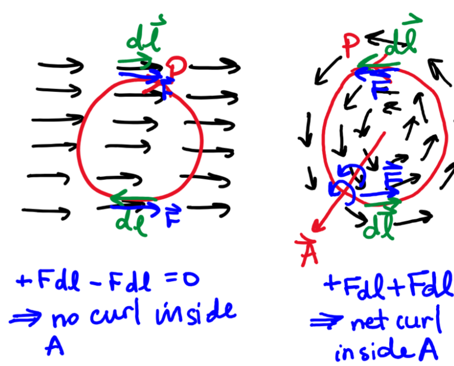
$$\underbrace{\int_V (\vec{\nabla} \cdot \vec{F})\, d\tau}_{\text{how field spreads inside the volume}} = \underbrace{\oint_S \vec{F} \cdot d\vec{a}}_{\text{flux of field through the surface}}$$



Stokes Theorem

States that inside a surface A bounded by a closed curve P, if a vector field $\vec{F}$ has a non-zero curl *about* $\vec{A}$ *(the normal to the surface)*, then there will be a non-zero circulation of $\vec{F}$ around the curve P.

$$\underbrace{\int_A (\vec{\nabla} \times \vec{F}) \cdot d\vec{a}}_{\text{flux of the curl of } \vec{F}} = \underbrace{\oint_P \vec{F} \cdot d\vec{l}}_{\text{circulation of } \vec{F}}$$



Key Electricity and Magnetism Laws

- **Coulomb's Law**

$$\vec{F}_e = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

- **Electric Field of a Point Charge**
  sets up a diverging field

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} \hat{r}$$

- **Magnetic Force**

$$\vec{F}_b = q\vec{v} \times \vec{B}$$

- **Biot-Savart Law**
  sets up a curling field

$$\vec{B} = \int \frac{\mu_0}{4\pi} \frac{i\, d\vec{s} \times \hat{r}}{r^2}$$

Unit 1: Electromagnetic Waves

Maxwell's Equations

1. **Gauss's Law for Electricity**
   Relates the electric flux through a closed 3D Gaussian surface to the total charge enclosed within that surface.
   $$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc}}}{\varepsilon_0}$$
   There exists a diverging electric field if there exists a non-zero charge density at that point.
   $$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

2. **Gauss's Law for Magnetism**
   Any 3D Gaussian surface will have zero net magnetic flux (no magnetic monopoles).
   $$\oint_S \vec{B} \cdot d\vec{a} = 0$$
   There exists no diverging magnetic field at any point in space because there are no magnetic monopoles.
   $$\vec{\nabla} \cdot \vec{B} = 0$$

3. **Faraday's Law of Induction**
   Relates the electric circulation around a closed Faradian loop to the changing magnetic flux through the surface bounded by the loop.
   $$\oint_P \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int_S \vec{B} \cdot d\vec{a}$$
   <u>Note:</u> For coils with N turns, multiply the flux by N.

   A changing magnetic field induces a circulating (curling) electric field.
   $$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

4. **Ampere-Maxwell Law**
   Relates the magnetic circulation around a closed Amperian loop to the enclosed current and changing electric flux through the surface bounded by the loop.
   $$\oint_P \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}, \quad \Phi_E = \int_S \vec{E} \cdot d\vec{a}$$
   Both a changing electric field and a non-zero current density induce a circulating (curling) magnetic field.
   $$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}, \quad I_{\text{enc}} = \int_S \vec{J} \cdot d\vec{a}$$

Deriving the Differential Forms of Maxwell's Equations

By applying the divergence and stokes theorems to the integral forms of Maxwell's equations, we can derive their corresponding differential forms. This process allows us to express the behavior of electric and magnetic fields locally at every point in space and time, rather than just over closed surfaces or loops.

1. **Gauss's Law for Electricity**

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{\text{enc}}}{\varepsilon_0}$$

Applying the divergence theorem to the left-hand side:

$$\oint_S \vec{E} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{E})\, d\tau$$

We can also write the enclosed charge as a volume integral of the charge density:

$$q_{\text{enc}} = \int_V \rho\, d\tau \qquad \rho = (\text{vol charge density})$$

Thus,

$$\implies \int_V (\vec{\nabla} \cdot \vec{E})\, d\tau = \frac{1}{\varepsilon_0} \int_V \rho\, d\tau$$

$$\implies \boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}}$$

2. **Gauss's Law for Magnetism**

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

Applying the divergence theorem:

$$\oint_S \vec{B} \cdot d\vec{a} = \int_V (\vec{\nabla} \cdot \vec{B})\, d\tau = 0$$

Thus,

$$\implies \boxed{\vec{\nabla} \cdot \vec{B} = 0}$$

3. **Faraday's Law of Induction**

$$\oint_P \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int_S \vec{B} \cdot d\vec{a}$$

Applying Stokes' theorem to the left-hand side:

$$\oint_P \vec{E} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a}$$

We can also write the changing magnetic flux as:

$$-\frac{d\Phi_B}{dt} = -\frac{d}{dt}\left[\int_S \vec{B}\cdot\, d\vec{a}\right] = -\int_S \frac{\partial \vec{B}}{\partial t}\cdot d\vec{a}$$

Note: Stokes theorem relates the circulation around a curve to the curl over a surface at a single instant in time. In other words, the surface S is *fixed* and does not change in time, so $d\vec{a}$ is *time independent* and the time derivative acts only on the integrand, allowing us to move the time derivative inside the integral.

Thus, equating the two sides gives:

$$\implies \int_S (\vec{\nabla}\times\vec{E})\cdot d\vec{a} = -\int_S \frac{\partial \vec{B}}{\partial t}\cdot d\vec{a}$$

$$\implies \boxed{\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}}$$

4. **Ampere-Maxwell Law**

$$\oint_P \vec{B}\cdot d\vec{l} = \mu_0 I_{\text{enc}} + \mu_0\varepsilon_0\frac{d\Phi_E}{dt}, \quad \Phi_E = \int_S \vec{E}\cdot\, d\vec{a}$$

Applying Stokes' theorem to the left-hand side:

$$\oint_P \vec{B}\cdot d\vec{l} = \int_S (\vec{\nabla}\times\vec{B})\cdot d\vec{a}$$

We can also write the enclosed current as a surface integral of the current density:

$$I_{\text{enc}} = \int_S \vec{J}\cdot d\vec{a}, \quad \vec{J} = (\text{area current density})$$

And the changing electric flux as:

$$\frac{d\Phi_E}{dt} = \frac{d}{dt}\left[\int_S \vec{E}\cdot\, d\vec{a}\right] = \int_S \frac{\partial \vec{E}}{\partial t}\cdot d\vec{a}$$

Thus, equating the two sides gives:

$$\implies \int_S (\vec{\nabla}\times\vec{B})\cdot d\vec{a} = \int_S \left(\mu_0\vec{J} + \mu_0\varepsilon_0\frac{\partial \vec{E}}{\partial t}\right)\cdot d\vec{a}$$

$$\implies \boxed{\vec{\nabla}\times\vec{B} = \mu_0\vec{J} + \mu_0\varepsilon_0\frac{\partial \vec{E}}{\partial t}}$$

Applying Vector Identities
Additionally, we can apply the vector identities we have to the differential forms of Maxwell's equations to derive some useful results.

Applying divergence of a curl is zero to Faraday's law:

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{E}) = 0$$
$$\vec{\nabla} \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) = 0$$
$$-\frac{\partial}{\partial t} \underbrace{(\nabla \cdot \vec{B})}_{\text{Gauss's law for magnetism } (=0)} = 0$$
$$\implies 0 = 0$$

Applying divergence of a curl is zero to Ampere-Maxwell law:

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) = 0$$
$$\vec{\nabla} \cdot \left(\mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) = 0$$
$$\mu_0 (\vec{\nabla} \cdot \vec{J}) + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \underbrace{(\nabla \cdot \vec{E})}_{\text{Gauss's law for electricity} (=\frac{\rho}{\varepsilon_0})} = 0$$
$$\mu_0 (\vec{\nabla} \cdot \vec{J}) + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(\frac{\rho}{\varepsilon_0} \right) = 0$$
$$\implies \boxed{\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0} \quad \text{(Continuity Equation)}$$

We can rewrite the continuity equation into an integral form:

Integrating both sides over a volume V:

$$\int_V (\vec{\nabla} \cdot \vec{J}) \, d\tau = \int_V -\frac{\partial \rho}{\partial t} \, d\tau$$

Apply divergence theorem to the left-hand side:

$$\int_V (\vec{\nabla} \cdot \vec{J}) \, d\tau = \oint_S \vec{J} \cdot d\vec{a}$$

Since the volume V is fixed in space, we can pull the time derivative outside the integral. We also know that the total charge enclosed is $q = \int_V \rho \, d\tau$:

$$\int_V -\frac{\partial \rho}{\partial t} \, d\tau = -\frac{d}{dt} \underbrace{\int_V \rho \, d\tau}_{q_{\text{enclosed}}} = -\frac{dq}{dt}$$

Thus, equating the two sides gives:

$$\boxed{\oint_S \vec{J} \cdot d\vec{a} = -\frac{dq}{dt}}$$

<u>Note:</u> The continuity equation is also called a flux based conservation law. It represents a local **conservation of charge** and states that if the total charge enclosed in a volume is changing ($\frac{dq}{dt} \neq 0$), then there must be a net current flowing into or out of the volume ($\oint_S \vec{J} \cdot d\vec{a} \neq 0$) that accounts for that change.

If a balloon were filled with a certain amount of water and that quantity of water was decreasing...

It must mean that there is a flux of water passing through the surface, leaving the volume!

Showing that E and B Fields Exist as Waves in a Vacuum
In a vacuum, there are no charges or currents, so we can set $\rho = 0$ and $\vec{J} = \vec{0}$ in Maxwell's equations. This simplifies them to:

$\vec{\nabla} \cdot \vec{E} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$	no divergent fields (no source charge)
$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$	only exist curling fields induced by one another!

By applying the curl of a curl rule to Faraday's and Ampere-Maxwell laws, we can derive the wave equations for the electric and magnetic fields in a vacuum.

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla}(\underbrace{\vec{\nabla} \cdot \vec{E}}_{0 \text{ in a vacuum}}) - \nabla^2 \vec{E}$$

$$\vec{\nabla} \times \left(-\frac{\partial \vec{B}}{\partial t}\right) = -\nabla^2 \vec{E}$$

$$-\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B}) = -\nabla^2 \vec{E}$$

$$-\frac{\partial}{\partial t}\left(\mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right) = -\nabla^2 \vec{E}$$

$$\Longrightarrow \boxed{\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}}$$

$$\vec{\nabla} \times (\underbrace{\vec{\nabla} \times \vec{B}}_{\mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \text{ in a vacuum}}) = \vec{\nabla}(\underbrace{\vec{\nabla} \cdot \vec{B}}_{0}) - \nabla^2 \vec{B}$$

$$\mu_0 \varepsilon_0 \frac{\partial}{\partial t}(\underbrace{\vec{\nabla} \times \vec{E}}_{-\frac{\partial \vec{B}}{\partial t}}) = -\nabla^2 \vec{B}$$

$$-\mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} = -\nabla^2 \vec{B}$$

$$\Longrightarrow \boxed{\nabla^2 \vec{B} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}}$$

<u>Recall:</u> The classic wave equation is

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad \Longrightarrow \quad v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = c$$

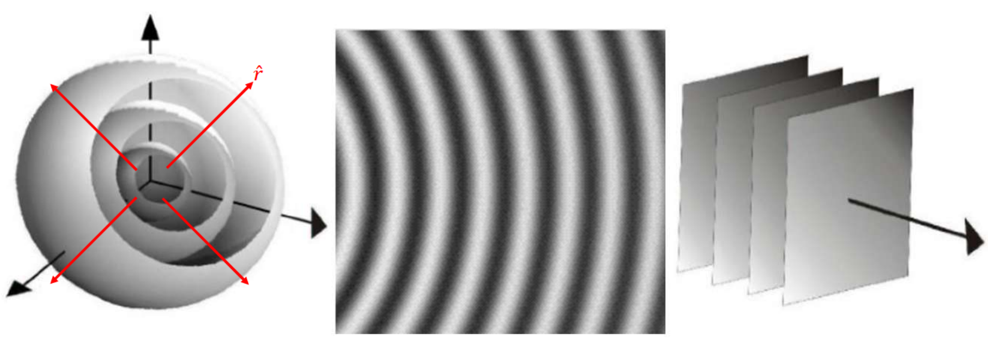
Modeling EM Radiation (Light) in a Vacuum

Properties of light:

- $\vec{E} \times \vec{B}$ points in the direction of propagation.
- $\vec{E}$ and $\vec{B}$ are perpendicular to each other and to the direction of propagation (transverse wave).
- $\vec{E}$ and $\vec{B}$ oscillate in phase as to perpetually induce each other according to Faraday's and Ampere-Maxwell laws.
- The magnitudes of $\vec{E}$ and $\vec{B}$ are related by $E = cB$.
- Light does not require a medium to propagate.
- Light has speed
$$v = c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

Our goal is to find a solution for $\vec{E}$ and $\vec{B}$ that satisfies the 3D wave equations as well as Maxwell's equations in a vacuum.

Plane Wave Approximation
Consider the following wavefronts generated by a point source of light (ie. a star):



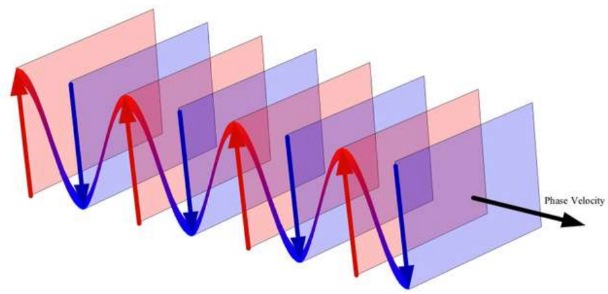
Close to the star, spherical wavefronts emanate radially outward as 3D shells

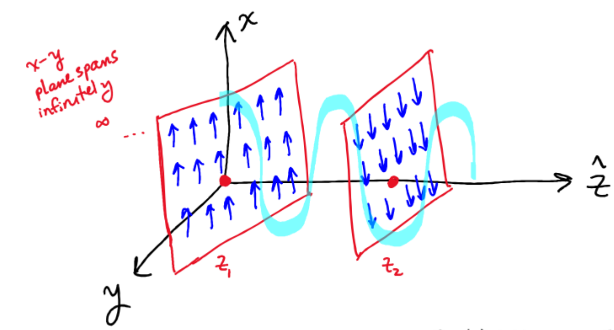
Very far from the star, wavefronts appear more like planes propagating in one direction

The radius of curvature of the wavefronts increases as they travel outward. Thus, at a far enough distance from the source, the wavefronts will appear to be flat planes consisting of electric and magnetic fields with *uniform* magnitude and direction. This is called the **plane wave approximation**.

The plane wave model also assumes **monochromatic light**, having a fixed wavelength λ and wavenumber k ($k = 2\pi/\lambda$). This also implies a fixed frequency ω, since the speed of light is constant:
$$v = \frac{\omega}{k} = c \quad \implies \quad \omega = ck.$$

Without a single frequency, the wavefronts would interfere with one another and would no longer form the ideal uniform planes described by the plane wave approximation.



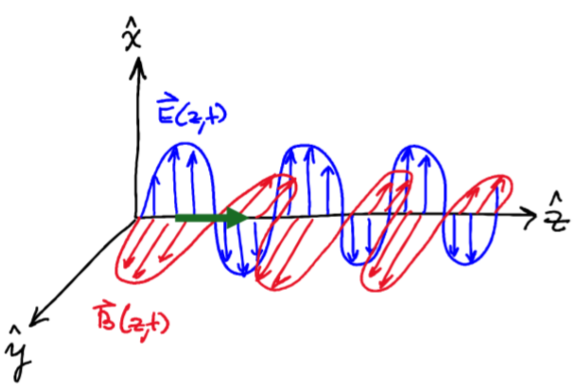


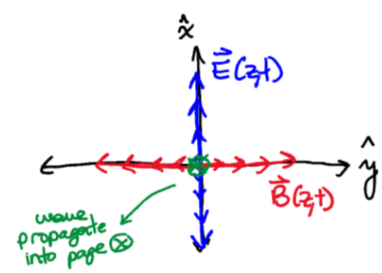
Note: Although light consists of both electric and magnetic fields, we will primarily focus on the electric field when modeling light, since the magnetic field can be readily derived from it using Faraday's law. We will also focus on the electric field when analyzing polarization, since the direction of the electric field determines the polarization of light.

Case $\hat{z}$

For the case $\hat{z}$, we take the direction of propagation to be along the $+z$ axis. At any fixed position z, the electric field $\vec{E}$ has a uniform magnitude and direction across the entire xy plane. As with the 1D plane wave, the *local view* at a fixed point along the z axis shows the electric field oscillating in time as the light wave passes through, with the electric field remaining transverse to the direction of propagation.

Note: Don't forget that the magnetic field $\vec{B}$ is also oscillating in phase with and is perpendicular to both $\vec{E}$ and the direction of propagation.





Therefore, a solution for the electric field of a plane wave in case $\hat{z}$ is:

$$\boxed{\vec{E} = \underbrace{\left(E_{0_x} e^{i\phi_x}\hat{x} + E_{0_y} e^{i\phi_y}\hat{y}\right)}_{\text{complex amplitude vector}} e^{i(kz-\omega t)}}$$

From the equation, we can see that the magnitudes of the electric field in the x and y directions need not be the same, and that there is no z component of the electric field, since the wave is transverse. Like the 1D plane wave, the space and time dependence of this 3D plane wave is contained in the phase $(kz - \omega t)$.

The amplitude of the electric field is given by

$$|\vec{E}| = \sqrt{|E_{0_x}|^2 + |E_{0_y}|^2}$$

Note: The phases ϕ_x and ϕ_y of the complex amplitude vector do not affect the magnitude of the electric field, but they do govern the polarization of the light wave. This is because of the dot product:

$$|\vec{E}|^2 = \vec{E} \cdot \vec{E}^*, \quad \vec{E}^* = (\text{complex conjugate of } \vec{E})$$

$$\vec{E} \cdot \vec{E}^* = \left(E_{0_x} e^{i\phi_x} \hat{x} + E_{0_y} e^{i\phi_y} \hat{y}\right) \cdot \left(E_{0_x} e^{-i\phi_x} \hat{x} + E_{0_y} e^{-i\phi_y} \hat{y}\right) = |E_{0_x}|^2 + |E_{0_y}|^2$$

The exponentials multiply to 1 (e^0), and $\hat{x} \cdot \hat{x} = \hat{y} \cdot \hat{y} = 1$, while $\hat{x} \cdot \hat{y} = \hat{y} \cdot \hat{x} = 0$, so the cross terms cancel out.

Note: The xy components for both the electric and magnetic fields have the same harmonic properties of k and ω (wavelength and frequency), since they are both derived from the same phase $(kz - \omega t)$.

Finally, note that we have switched from sinusoidal functions to complex exponentials. This is because complex exponentials are easier to work with mathematically, and if we want to get back to the real-valued (physically measurable) electric field, we can simply take the real part of the complex expression (using Euler's formula).

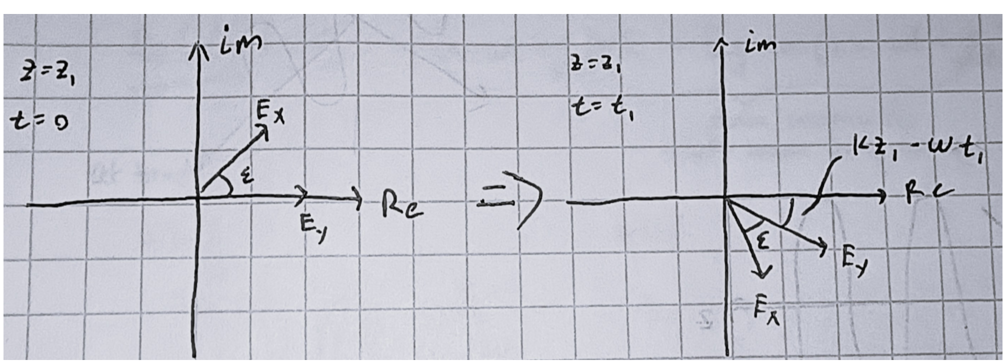
$$\vec{E}_{\text{real}} = \Re\Big\{\vec{E}\Big\} = \Re\Big\{\left(E_{0_x} e^{i\phi_x} \hat{x} + E_{0_y} e^{i\phi_y} \hat{y}\right) e^{i(kz - \omega t)}\Big\}$$

$$\vec{E}_{\text{real}} = E_{0_x} \cos(kz - \omega t + \phi_x)\, \hat{x} + E_{0_y} \cos(kz - \omega t + \phi_y)\, \hat{y}$$

Polarization of Light

Polarization has been mentioned a few times already, but what exactly is it? Simply put, polarization describes the path traced out by the tip of the electric field vector $\vec{E}$ in the xy plane at a fixed z as the light wave passes through it. How can we visualize this? With *parametric equations.*

First, to better understand the phase difference between the x and y components of the electric field, we draw a phasor diagram in the complex plane:



Where

$$E_x = E_{0_x} e^{i(\phi_x + kz_1 - \omega t)}$$
$$E_y = E_{0_y} e^{i(\phi_y + kz_1 - \omega t)}$$
$$\boxed{\varepsilon = \phi_x - \phi_y \quad \text{(phase difference)}}$$

Since $z = z_1$ is fixed, kz_1 becomes a constant phase offset, so the rotation of the phasors is driven entirely by $-\omega t$ (CW, since it is negative). The phase $kz_1 - \omega t$ is the same for both E_x and E_y, so the relative phase difference ε between the two components remains constant as the light wave propagates. Also note that while the magnitudes of E_x and E_y are constant in the phasor diagram, their *real* magnitudes are given by the projections of the phasors onto the real axis, which oscillate in time (allowing elliptical polarization, where the magnitude of $\vec{E}$ varies in time).

General Polarization

The most general case of polarization is elliptical polarization, where linear and circular polarization are just degenerate ellipses. To see this, we will rewrite the *real* electric field components in terms of the phase difference ε:

$$\vec{E}_{\text{real}} = E_{0_x} \cos(kz_1 - \omega t + \phi_x)\,\hat{x} + E_{0_y} \cos(kz_1 - \omega t + \phi_y)\,\hat{y}$$
$$x(t) = E_{0_x} \cos(\omega t - \varepsilon)$$
$$y(t) = E_{0_y} \cos(\omega t)$$

Note: kz_1 is constant, so it cancels out when we take the difference of the two phases, but we keep ωt in the argument since it is the time-varying part. Because we define ε to be positive, it appears in the x component (which is why E_x leads E_y in the phasor diagram). The argument of the cosine is $\varepsilon - \omega t$, but since cosine is an even function, we can equivalently write it as $\omega t - \varepsilon$.

We now deparametrize the equations to obtain a single equation in x and y that describes the path traced out by $\vec{E}$ in the xy plane. First expanding the cosine using sum and difference:

$$x(t) = E_{0_x}(\cos(\omega t)\cos(\varepsilon) + \sin(\omega t)\sin(\varepsilon))$$
$$y(t) = E_{0_y} \cos(\omega t) \implies \cos(\omega t) = \frac{y}{E_{0_y}}$$

Substituting $\cos(\omega t)$ into the x equation:

$$x(t) = E_{0_x}\left(\frac{y}{E_{0_y}}\cos(\varepsilon) + \sin(\omega t)\sin(\varepsilon)\right)$$

Solving for $\sin(\omega t)$:

$$\sin(\omega t) = \frac{1}{\sin(\varepsilon)}\left(\frac{x}{E_{0_x}} - \frac{y}{E_{0_y}}\cos(\varepsilon)\right)$$

Plugging $\sin(\omega t)$ and $\cos(\omega t)$ into the Pythagorean identity $\cos^2(\omega t) + \sin^2(\omega t) = 1$:

$$\left(\frac{y}{E_{0_y}}\right)^2 + \left(\frac{1}{\sin(\varepsilon)}\left(\frac{x}{E_{0_x}} - \frac{y}{E_{0_y}}\cos(\varepsilon)\right)\right)^2 = 1$$

Simplify:

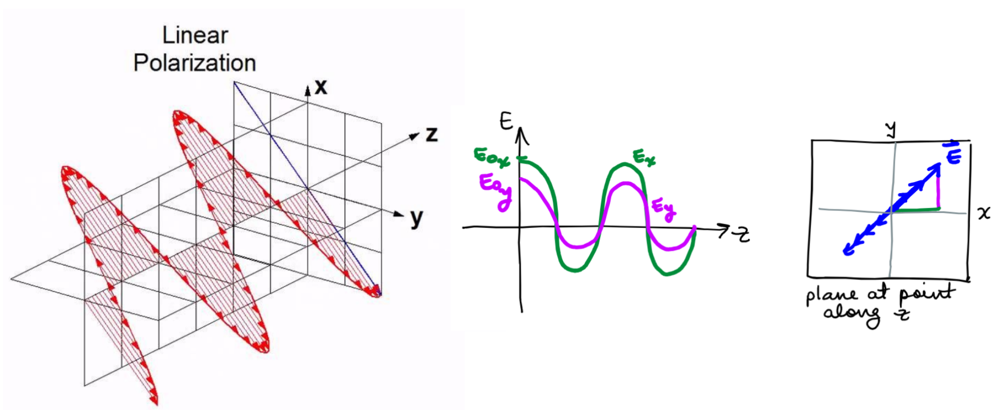
$$\boxed{\frac{x^2}{E_{0_x}^2}+\frac{y^2}{E_{0_y}^2}-2\frac{xy}{E_{0_x}E_{0_y}}\cos(\varepsilon)=\sin^2(\varepsilon)}$$

Note: This is the form of an ellipse centered at the origin ($ax^2+bxy+cy^2=d$)

Therefore, for any ε that doesn't result in a degenerate case, the polarization of light is elliptical. Now we will list some cases so we don't have to keep referring back to the general equation:

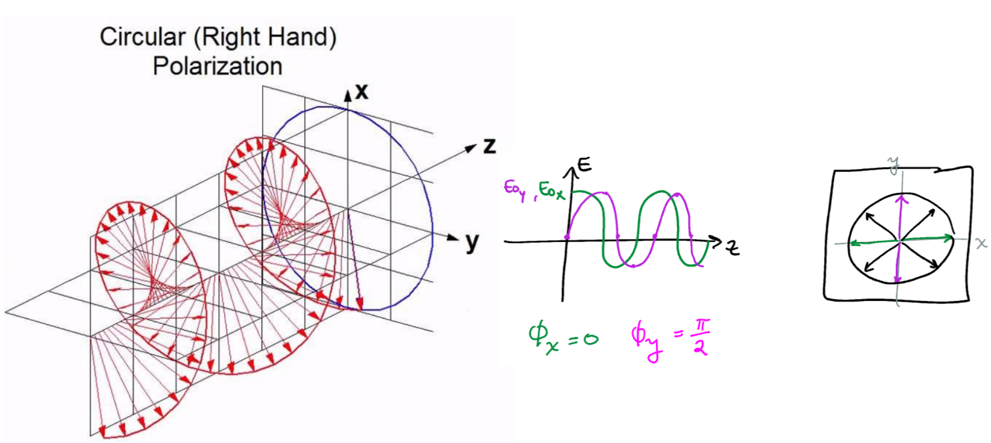
- **Linear Polarization:** E_x and E_y are in phase and the ratio of their magnitudes is constant (slope of the line).

$$\boxed{\varepsilon=n\pi,\quad n=(0,1,2,\dots)}$$



- **Circular Polarization:** E_x and E_y have equal magnitudes and are $\pi/2$ out of phase.

$$\boxed{E_{0_x}=E_{0_y},\quad \text{and}\quad \varepsilon=(2n+1)\frac{\pi}{2},\quad n=(0,1,2,\dots)}$$

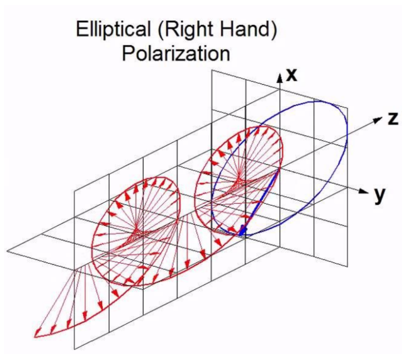


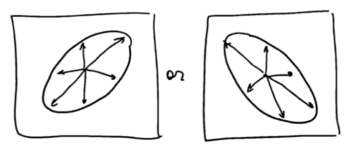
- **Elliptical Polarization:** All other cases

$$\boxed{E_{0_x} \neq E_{0_y}, \quad \text{and} \quad \varepsilon \neq n\pi \quad \text{(otherwise it will be linear)}}$$

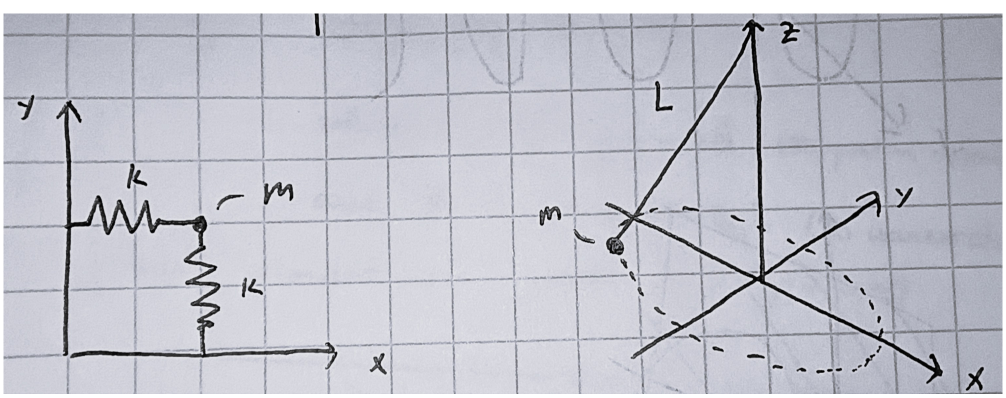
OR

$$\boxed{E_{0_x} = E_{0_y}, \quad \text{and} \quad \varepsilon \neq (2n+1)\frac{\pi}{2} \quad \text{(otherwise it will be circular)}}$$





Why Ellipses?
Fundamentally, the electric and magnetic fields behave like oscillators, so the path traced out by the electric field vector must be periodic and closed. The only conic section that is both periodic and closed is an ellipse, with circles and lines appearing as degenerate cases. Another way to visualize this is to think of E_x and E_y as two simple harmonic oscillators (SHOs) that are coupled through a common driving frequency, analogous to two springs with equal spring constant k attached to the same mass. The motion of the mass is then a superposition of the two SHOs, and the resulting path traced out by the mass is an ellipse. Feynman also introduces the picture of a string-mass pendulum, where the mass can swing in any direction in the xy plane, and the path traced out by the mass is likewise an ellipse. This intuition is useful because it emphasizes that the driving phase ωt is the same for both the x and y components, and that the phase difference, and thus the resulting trace, depends only on the initial orientation of the motion.



General Plane Wave Solution

We would like to generalize the case $\hat{z}$ solution

$$\vec{E} = \left(E_{0_x} e^{i\phi_x} \hat{x} + E_{0_y} e^{i\phi_y} \hat{y}\right) e^{i(kz - \omega t)}$$

to describe a plane wave propagating in an arbitrary direction. To do this, we can simply replace the kz term in the phase with $\vec{k} \cdot \vec{r}$, where $\vec{k}$ is the **wavevector** that points in the direction of propagation and has magnitude k. This allows us to describe plane waves propagating in any direction, not just along the z axis.

$$\vec{k} = k_x \hat{x} + k_y \hat{y} + k_z \hat{z}$$

$$|\vec{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2} = k, \quad k = \frac{2\pi}{\lambda}$$

and $\vec{r}$ is the position vector from to origin to any point in space:

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

Therefore, the general plane wave solution is:

$$\boxed{\vec{E} = \left(E_{0_x} e^{i\phi_x} \hat{x} + E_{0_y} e^{i\phi_y} \hat{y}\right) e^{i(\vec{k} \cdot \vec{r} - \omega t)}}$$

This general form of the plane wave solution is intuitive because it is consistent with our understanding that $\vec{E}$ depends only on spatial variations along the direction of propagation. The dot product $\vec{k} \cdot \vec{r}$ extracts the component of $\vec{r}$ in the direction of $\vec{k}$, which for case $\hat{z}$ reduces to variations along the z axis. At the same time, $\vec{E}$ does not depend on spatial variations perpendicular to the direction of propagation (it is a transverse wave). For any $\vec{r}$ perpendicular to $\vec{k}$, we have $\vec{k} \cdot \vec{r} = 0$, so in case $\hat{z}$ there are no variations in the x and y directions.

Useful Properties of the General Plane Wave Form

Time derivative

$$\frac{\partial \vec{E}}{\partial t} = -i\omega \vec{E}, \quad \frac{\partial^2 \vec{E}}{\partial t^2} = -\omega^2 \vec{E}$$

Spatial derivative

$$\vec{\nabla} \cdot \vec{E} = i\vec{k} \cdot \vec{E}, \quad \nabla^2 \vec{E} = -k^2 \vec{E}$$

$$\vec{\nabla} \times \vec{E} = i\vec{k} \times \vec{E}$$

<u>Note:</u> These properties follow from the fact that the only spatial and temporal dependence of $\vec{E}$ is contained in the phase $(\vec{k} \cdot \vec{r} - \omega t)$. When taking derivatives, the complex amplitude vector can therefore be treated as a constant, and the chain rule applies directly to the exponential term.

Also since we can find the direction of the magnetic field $\vec{B}$ using Faraday's law ($\vec{\nabla} \times \vec{E} = i\vec{k} \times \vec{E}$) and the relationship between the magnitudes of $\vec{E}$ and $\vec{B}$ ($|\vec{E}| = c|\vec{B}|$), we can also write:

$$\boxed{\vec{B} = \frac{1}{c} \hat{k} \times \vec{E}}$$

Constraints on Plane Wave Solutions from Maxwell's Equations
For **case $\hat{\mathbf{z}}$**, we have the general plane wave solution of the form

$$\vec{E} = E_x\hat{x} + E_y\hat{y} + E_z\hat{z}$$

$$\vec{B} = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}$$

Where E_x, E_y, E_z, B_x, B_y, and B_z are complex functions of the form:

$$E_x = E_{0_x}e^{i(kz-\omega t+\phi_{x_E})}$$

Using the properties of time and spatial derivatives of the general plane wave form, we can apply Maxwell's equations to derive constraints on the components of $\vec{E}$ and $\vec{B}$.

(1) Applying Gauss's law for electricity:

$$\vec{\nabla}\cdot\vec{E} = 0$$

$$\left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\right) = 0$$

Since E_x and E_y are only dependent on z and t, their spatial derivatives with respect to x and y are zero.

$$\frac{\partial E_x}{\partial x} = \frac{\partial E_y}{\partial y} = 0$$

Thus,

$$\frac{\partial E_z}{\partial z} = \frac{\partial}{\partial z}\left(E_{0_z}e^{i(kz-\omega t+\phi_{z_E})}\right) = 0$$

$$\implies ikE_{0_z}e^{i(kz-\omega t+\phi_{z_E})} = 0$$

Since the exponential term is never zero, $\boxed{E_{0_z} \text{ must be zero}}$.

This matches our expectation that the electric field of a plane wave must be transverse to the direction of propagation, so there can be no z component of the electric field for case $\hat{z}$.

(2) Applying Gauss's law for magnetism:

$$\vec{\nabla}\cdot\vec{B} = 0$$

Similarly to (1), we can see that B_{0_z} must also be zero. Also note that the lack of an x and y dependence in both $\vec{E}$ and $\vec{B}$ shows that the fields are *uniform* across the entire xy plane, which is consistent with the plane wave approximation.

(3) Applying Faraday's law: Setting E_{0_z} and B_{0_z} to zero as found in (1) and (2).

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & 0 \end{vmatrix} = -\frac{\partial}{\partial z}E_y\hat{x} + \frac{\partial}{\partial z}E_x\hat{y} + \underbrace{\left(\frac{\partial}{\partial x}E_y - \frac{\partial}{\partial y}E_x\right)}_{0}\hat{z}$$

$$\begin{aligned}
&= -\frac{\partial}{\partial z}\left(E_{0_y}e^{i(kz-\omega t+\phi_{y_E})}\right)\hat{x} + \frac{\partial}{\partial z}\left(E_{0_x}e^{i(kz-\omega t+\phi_{x_E})}\right)\hat{y} \\
&= -ikE_{0_y}e^{i(kz-\omega t+\phi_{y_E})}\hat{x} + ikE_{0_x}e^{i(kz-\omega t+\phi_{x_E})}\hat{y} \\
&= ike^{i(kz-\omega t)}\left(-E_{0_y}e^{i\phi_{y_E}}\hat{x} + E_{0_x}e^{i\phi_{x_E}}\hat{y}\right)
\end{aligned}$$

Now solving for $-\frac{\partial \vec{B}}{\partial t}$.

$$\begin{aligned}
-\frac{\partial}{\partial t}\left(B_x\hat{x} + B_y\hat{y}\right) &= -\left[\frac{\partial}{\partial t}\left(B_{0_x}e^{i(kz-\omega t+\phi_{x_B})}\right)\hat{x} + \frac{\partial}{\partial t}\left(B_{0_y}e^{i(kz-\omega t+\phi_{y_B})}\right)\hat{y}\right] \\
&= -\left[-i\omega B_{0_x}e^{i(kz-\omega t+\phi_{x_B})}\hat{x} - i\omega B_{0_y}e^{i(kz-\omega t+\phi_{y_B})}\hat{y}\right] \\
&= i\omega e^{i(kz-\omega t)}\left(B_{0_x}e^{i\phi_{x_B}}\hat{x} + B_{0_y}e^{i\phi_{y_B}}\hat{y}\right)
\end{aligned}$$

Setting them equal to each other gives:

$$k\left(-E_{0_y}e^{i\phi_{y_E}}\hat{x} + E_{0_x}e^{i\phi_{x_E}}\hat{y}\right) = \omega\left(B_{0_x}e^{i\phi_{x_B}}\hat{x} + B_{0_y}e^{i\phi_{y_B}}\hat{y}\right)$$

Looking at the x components, the magnitudes and phases must match:

$$\begin{aligned}
-kE_{0_y}e^{i\phi_{y_E}} &= \omega B_{0_x}e^{i\phi_{x_B}} \\
\implies E_{0_y}e^{i\pi}e^{i\phi_{y_E}} &= \frac{\omega}{k}B_{0_x}e^{i\phi_{x_B}}
\end{aligned}$$

Thus,

$$\boxed{\frac{E_{0_y}}{B_{0_x}} = \frac{\omega}{k} = c, \quad \phi_{y_E} + \pi = \phi_{x_B}}$$

Looking at the y components, the magnitudes and phases must match:

$$\begin{aligned}
kE_{0_x}e^{i\phi_{x_E}} &= \omega B_{0_y}e^{i\phi_{y_B}} \\
\implies E_{0_x}e^{i\phi_{x_E}} &= \frac{\omega}{k}B_{0_y}e^{i\phi_{y_B}}
\end{aligned}$$

Thus,

$$\boxed{\frac{E_{0_x}}{B_{0_y}} = \frac{\omega}{k} = c, \quad \phi_{x_E} = \phi_{y_B}}$$

<u>(4) Applying Ampere-Maxwell law:</u> Setting E_{0_z} and B_{0_z} to zero as found in (1) and (2).

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t}$$

We will see the same relationships found in (3) emerge, physics is consistent!

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & B_y & 0 \end{vmatrix} = -\frac{\partial}{\partial z}B_y\hat{x} + \frac{\partial}{\partial z}B_x\hat{y} + \underbrace{\left(\frac{\partial}{\partial x}B_y - \frac{\partial}{\partial y}B_x\right)}_{0}\hat{z}$$
$$= -\frac{\partial}{\partial z}\left(B_{0_y}e^{i(kx-\omega t+\phi_{y_B})}\right)\hat{x} + \frac{\partial}{\partial z}\left(B_{0_x}e^{i(kx-\omega t+\phi_{x_B})}\right)\hat{y}$$
$$= ike^{i(kx-\omega t)}\left[-B_{0_y}e^{i\phi_{y_B}}\hat{x} + B_{0_x}e^{i\phi_{x_B}}\hat{y}\right]$$

Now solving for $\frac{1}{c^2}\frac{\partial \vec{E}}{\partial t}$.

$$\frac{1}{c^2}\frac{\partial}{\partial t}(E_x\hat{x} + E_y\hat{y}) = \frac{1}{c^2}\left[\frac{\partial}{\partial t}(E_{0_x}e^{i(kx-\omega t+\phi_{x_E})})\hat{x} + \frac{\partial}{\partial t}(E_{0_y}e^{i(kx-\omega t+\phi_{y_E})})\hat{y}\right]$$
$$= \frac{1}{c^2}\left[-i\omega E_{0_x}e^{i(kx-\omega t+\phi_{x_E})}\hat{x} - i\omega E_{0_y}e^{i(kx-\omega t+\phi_{y_E})}\hat{y}\right]$$
$$= -\frac{1}{c^2}i\omega e^{i(kx-\omega t)}\left[E_{0_x}e^{i\phi_{x_E}}\hat{x} + E_{0_y}e^{i\phi_{y_E}}\hat{y}\right]$$

$\hat{x}$ components:

$$-kB_{0_y}e^{i\phi_{y_B}} = -\frac{1}{c^2}\omega E_{0_x}e^{i\phi_{x_E}}$$
$$\implies B_{0_y}e^{i\phi_{y_B}} = \frac{1}{c^2}\frac{\omega}{k}E_{0_x}e^{i\phi_{x_E}} \qquad \implies \boxed{B_{0_y} = \frac{1}{c}E_{0_x}, \quad \phi_{y_B} = \phi_{x_E}}$$

$\hat{y}$ components:

$$kB_{0_x}e^{i\phi_{x_B}} = -\frac{1}{c^2}\omega E_{0_y}e^{i\phi_{y_E}}$$
$$\implies B_{0_x}e^{i\phi_{x_B}} = \frac{1}{c^2}\frac{\omega}{k}E_{0_y}e^{i\pi}e^{i\phi_{y_E}} \qquad \implies \boxed{B_{0_x} = \frac{1}{c}E_{0_y}, \quad \phi_{x_B} = \phi_{y_E} + \pi}$$

A *slightly* better method

Regrettably, while this previous method is rigorous, it is also very tedious and didn't use our newfound properties of general plane wave derivatives! Below is a (quick, I hope) method from Professor Corn:

We saw from before that when applying Faraday's law, we get

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$-\frac{\partial E_y}{\partial z}\hat{x} + \frac{\partial E_x}{\partial z}\hat{y} = -\frac{\partial B_x}{\partial t}\hat{x} - \frac{\partial B_y}{\partial t}\hat{y}$$

Setting the terms equal:

$$\implies \frac{\partial E_y}{\partial z} = \frac{\partial B_x}{\partial t} \quad \text{and} \quad \frac{\partial E_x}{\partial z} = -\frac{\partial B_y}{\partial t}$$

From the general plane wave solution:

$$\frac{\partial E_y}{\partial z} = \frac{\partial}{\partial z}\left[E_{0_y} e^{i\phi_{y_E}} e^{i(kz-\omega t)}\right] = ikE_y \quad \text{and} \quad \frac{\partial E_x}{\partial z} = ikE_x$$
$$\frac{\partial B_x}{\partial t} = \frac{\partial}{\partial t}\left[B_{0_x} e^{i\phi_{x_B}} e^{i(kz-\omega t)}\right] = -i\omega B_x \quad \text{and} \quad \frac{\partial B_y}{\partial t} = -i\omega B_y$$

$$\implies ikE_y = -i\omega B_x \qquad \implies ikE_x = -(-i\omega B_y)$$
$$\frac{E_y}{B_x} = \frac{-\omega}{k} = -c \qquad\qquad \frac{E_x}{B_y} = \frac{\omega}{k} = c$$

Plugging in for E_x, E_y, B_x, and B_y:

$$\frac{E_y}{B_x} = \frac{E_{0_y} e^{i\phi_{y_E}} e^{i(kz-\omega t)}}{B_{0_x} e^{i\phi_{x_B}} e^{i(kz-\omega t)}} = \frac{E_{0_y}}{B_{0_x}} e^{i(\phi_{y_E}-\phi_{x_B})} = -c, \quad -c = ce^{i\pi}$$

Equating the magnitudes and phases once again gives:

$$\frac{E_{0_y}}{B_{0_x}} = c \quad \text{and} \quad \phi_{y_E} - \phi_{x_B} = \pi$$

Similarly for the x component:

$$\frac{E_x}{B_y} = \frac{E_{0_x} e^{i\phi_{x_E}} e^{i(kz-\omega t)}}{B_{0_y} e^{i\phi_{y_B}} e^{i(kz-\omega t)}} = \frac{E_{0_x}}{B_{0_y}} e^{i(\phi_{x_E}-\phi_{y_B})} = c$$

Equating the magnitudes and phases once again gives:

$$\frac{E_{0_x}}{B_{0_y}} = c \quad \text{and} \quad \phi_{x_E} - \phi_{y_B} = 0$$

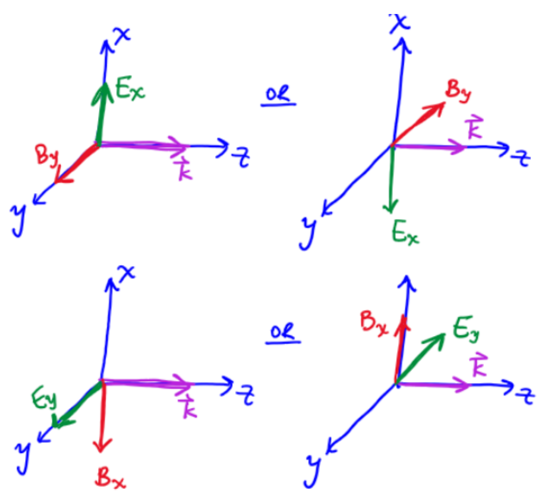
Note: If we didn't add in a phase difference of π then we would see that $E_{0_y}/B_{0_x} = -\omega/k$. In either case, the right-hand rule requires E_{0_y} and B_{0_x} to have opposite signs since $\vec{E}\times\vec{B}$ always points in the $\vec{k}$ direction. This can be expressed either by introducing a phase difference of π or by keeping the negative sign in the ratio.

Assuming $\vec{k}$ is in $+\hat{z}$, then either E_{0x} and B_{0y} are both positive or both negative!

$$\Rightarrow \frac{E_{0x}}{B_{0y}} = c$$

Assuming $\vec{k}$ is in $+\hat{z}$, then E_{0y} and B_{0x} must have opposite signs!

$$\Rightarrow \frac{E_{0y}}{B_{0x}} = -c$$



Energy Transported by EM Waves

Recall that energy is stored in the electric and magnetic fields, with energy densities given by

$$u_E = \frac{1}{2}\varepsilon_0 |\vec{E}|^2 \quad \text{(energy density of electric field)}$$

$$u_B = \frac{1}{2\mu_0}|\vec{B}|^2 \quad \text{(energy density of magnetic field)}$$

<u>Note:</u> As mentioned before, since we are now dealing with physically measurable quantities of the $\vec{E}$ and $\vec{B}$ fields, we need to use the *real* components of the fields.

Additionally, we are ignoring phase differences for now to simplify calculations, so $\phi_x = \phi_y = \phi_z = 0$ for the electric field (this simplification also applies to the magnetic field, but we'll focus on just the electric field for now).

$$\vec{E} = \Re\Big\{\vec{E}_{\text{complex}}\Big\} = \Re\Big\{\big(E_{0_x}\hat{x} + E_{0_y}\hat{y} + E_{0_z}\hat{z}\big)e^{i(\vec{k}\cdot\vec{r}-\omega t)}\Big\}$$

$$\vec{E} = E_{0_x}\cos(\vec{k}\cdot\vec{r}-\omega t)\,\hat{x} + E_{0_y}\cos(\vec{k}\cdot\vec{r}-\omega t)\,\hat{y} + E_{0_z}\cos(\vec{k}\cdot\vec{r}-\omega t)\,\hat{z}$$

Now the magnitude of the electric field is given by the dot product:

$$|\vec{E}|^2 = \vec{E}\cdot\vec{E} = |\vec{E}_0|^2\cos^2(\vec{k}\cdot\vec{r}-\omega t), \quad \text{where} \quad \vec{E}_0 = E_{0_x}\hat{x} + E_{0_y}\hat{y} + E_{0_z}\hat{z}$$

<u>For EM Waves</u>

- Since the magnitudes of the electric and magnetic fields are related by $|\vec{E}| = c|\vec{B}|$, the energy densities of the electric and magnetic fields are also equal ($c = \frac{1}{\sqrt{\varepsilon_0\mu_0}}$):

$$\boxed{u_E = u_B}$$

- The **total energy density** u is then the sum of the electric and magnetic energy densities:

$$\boxed{u = u_E + u_B = 2u_E = \varepsilon_0|\vec{E}|^2}$$

Poynting Vector $\vec{S}$ We can derive the Poynting vector $\vec{S}$, which describes the flow of electromagnetic energy carried by the $\vec{E}$ and $\vec{B}$ fields in the direction of $\vec{k}$. Its units are power per unit area, W/m^2. We can derive $\vec{S}$ by applying the *continuity equation* (local conservation of energy) to the energy density u. We should expect to see that:

$$\frac{\partial u}{\partial t} + \nabla\cdot\vec{S} = 0,$$

Similar to how we relate the change in charge density to the divergence of current density in the continuity equation for charge, here the change in energy density is related to the divergence of the Poynting vector which describes the rate at which energy flows through an area.

Starting from $u = \frac{1}{2}\varepsilon_0 \vec{E}\cdot\vec{E} + \frac{1}{2\mu_0}\vec{B}\cdot\vec{B}$, we want to show that $\frac{\partial u}{\partial t} = -\vec{\nabla}\cdot\vec{S}$:

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial t}\left[\frac{1}{2}\varepsilon_0 \vec{E}\cdot\vec{E} + \frac{1}{2\mu_0}\vec{B}\cdot\vec{B}\right] = \frac{1}{2}\varepsilon_0 \frac{\partial}{\partial t}(\vec{E}\cdot\vec{E}) + \frac{1}{2\mu_0}\frac{\partial}{\partial t}(\vec{B}\cdot\vec{B})$$

Applying the product rule to the dot products:

$$\frac{\partial}{\partial t}(\vec{E}\cdot\vec{E}) = 2\vec{E}\cdot\frac{\partial \vec{E}}{\partial t} \quad \text{and} \quad \frac{\partial}{\partial t}(\vec{B}\cdot\vec{B}) = 2\vec{B}\cdot\frac{\partial \vec{B}}{\partial t}$$

$$\implies \frac{\partial u}{\partial t} = \varepsilon_0 \vec{E}\cdot\frac{\partial \vec{E}}{\partial t} + \frac{1}{\mu_0}\vec{B}\cdot\frac{\partial \vec{B}}{\partial t}$$

From Faraday's Law:

$$\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \implies \quad \frac{\partial \vec{B}}{\partial t} = -\vec{\nabla}\times\vec{E}$$

From Ampere-Maxwell Law:

$$\vec{\nabla}\times\vec{B} = \mu_0\varepsilon_0\frac{\partial \vec{E}}{\partial t} \quad \implies \quad \frac{\partial \vec{E}}{\partial t} = \frac{1}{\mu_0\varepsilon_0}\vec{\nabla}\times\vec{B}$$

Substituting these into the expression for $\frac{\partial u}{\partial t}$:

$$\begin{aligned}\frac{\partial u}{\partial t} &= \varepsilon_0 \vec{E}\cdot\left(\frac{1}{\mu_0\varepsilon_0}\vec{\nabla}\times\vec{B}\right) + \frac{1}{\mu_0}\vec{B}\cdot(-\vec{\nabla}\times\vec{E}) \\ &= \frac{1}{\mu_0}\left[\vec{E}\cdot(\vec{\nabla}\times\vec{B}) - \vec{B}\cdot(\vec{\nabla}\times\vec{E})\right] \\ &= \frac{1}{\mu_0}\left[-\vec{\nabla}\cdot(\vec{E}\times\vec{B})\right]\end{aligned}$$

<u>Note:</u> The last step uses the vector identity $\vec{\nabla}\cdot(\vec{A}\times\vec{B}) = \vec{B}\cdot(\vec{\nabla}\times\vec{A}) - \vec{A}\cdot(\vec{\nabla}\times\vec{B})$.

Therefore, the Poynting vector is

$$\boxed{\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B}}$$

It is also equivalent to

$$\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B} = \frac{1}{\mu_0}|\vec{E}|\underbrace{|\vec{B}|}_{\frac{1}{c}|\vec{E}|}\hat{k} = \frac{1}{\underbrace{\mu_0 c}_{\varepsilon_0 c}}|\vec{E}|^2\hat{k}$$

$$\implies \boxed{\vec{S} = cu\hat{k}}$$

Intensity
Intensity is defined as the time-averaged magnitude of the Poynting vector, which describes the *average* power per unit area carried by the electromagnetic wave.

$$\langle \vec{S} \rangle = c\langle u \rangle \hat{k}, \qquad \langle \quad \rangle : \text{ time average}$$

Where

$$\langle u \rangle = \varepsilon_0 \langle |\vec{E}|^2 \rangle = \varepsilon_0 E_0^2 \langle \cos^2(\vec{k} \cdot \vec{r} - \omega t) \rangle = \frac{1}{2}\varepsilon_0 E_0^2$$

The time average of $\cos^2$ is $1/2$ because the function oscillates between 0 and 1, and over a full period, it spends equal time at all values in that range. Therefore, the average value of $\cos^2$ over one period is $1/2$.

Thus, the intensity of an electromagnetic wave is given by

$$\boxed{I = |\langle \vec{S} \rangle| = c\langle u \rangle = \frac{1}{2}\varepsilon_0 E_0^2 c}$$

Note: A higher light intensity corresponds to a larger electric-field amplitude, meaning the wave transports energy at a greater rate per unit area. This aligns with the physical interpretation of intensity as a measure of brightness: brighter light delivers more energy to a surface per unit time.

When dealing with electromagnetic waves, taking the time average is important because the fields oscillate rapidly, and we typically observe them on a macroscopic scale where the average behavior is more meaningful than the instantaneous values.

Unit 2: Radiation and Scattering

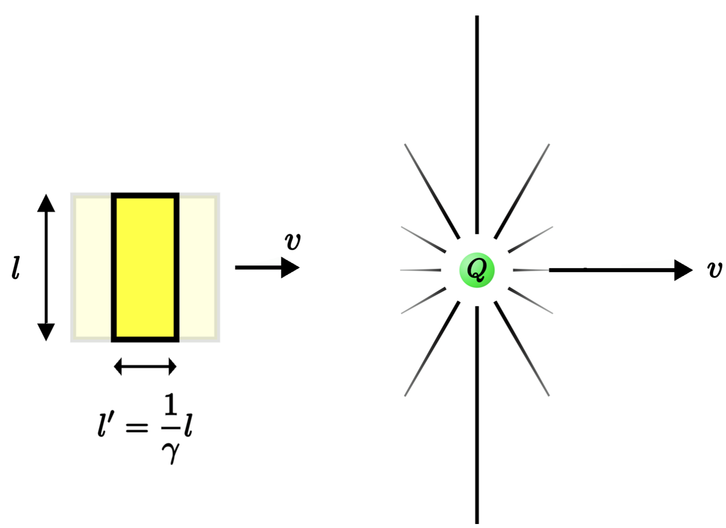
The goal of this unit is to model how electromagnetic waves are generated by an accelerated charge (via radiation and scattering) using *classical physics*. Along the way, we will make some approximations and assumptions that will yield a usable model but also introduce some limitations that we will later address with quantum mechanics.

Thompson Radiation

In the Thompson model, we are considering the electromagnetic radiation emitted by a *free* charged particle previously travelling at a constant velocity (or at rest) that is accelerated by a short *constant* acceleration, a "kick". This acceleration causes a kink in the electric field lines of the charge, which then propagates outward as an electromagnetic wave.

Before we begin, it is important to note the *scale* of the set-up we are working with. Perhaps the most important simplifying approximation we make is that the duration of the acceleration τ is very short compared to the time *after the acceleration* at which we observe the electric field, T ($\tau \ll T$). Additionally, τ is so short that the final constant velocity of the charge after the acceleration remains much less than the speed of light ($v \ll c$), so we can ignore relativistic effects. Importantly, this approximation lets us treat the electric field of a charge moving at constant velocity as essentially the same as that of a static charge, except that the field now points away from the *instantaneous position* of the charge rather than from a fixed point such as the origin.

For a charge moving relativistic velocity, something similar to length contraction occurs, where the electric field lines become squished along the direction of motion, and the field becomes stronger in the plane perpendicular to the motion and the lorentz factor γ appears. For our case, $v \ll c$, so $\gamma \approx 1$ and we can ignore these relativistic effects.



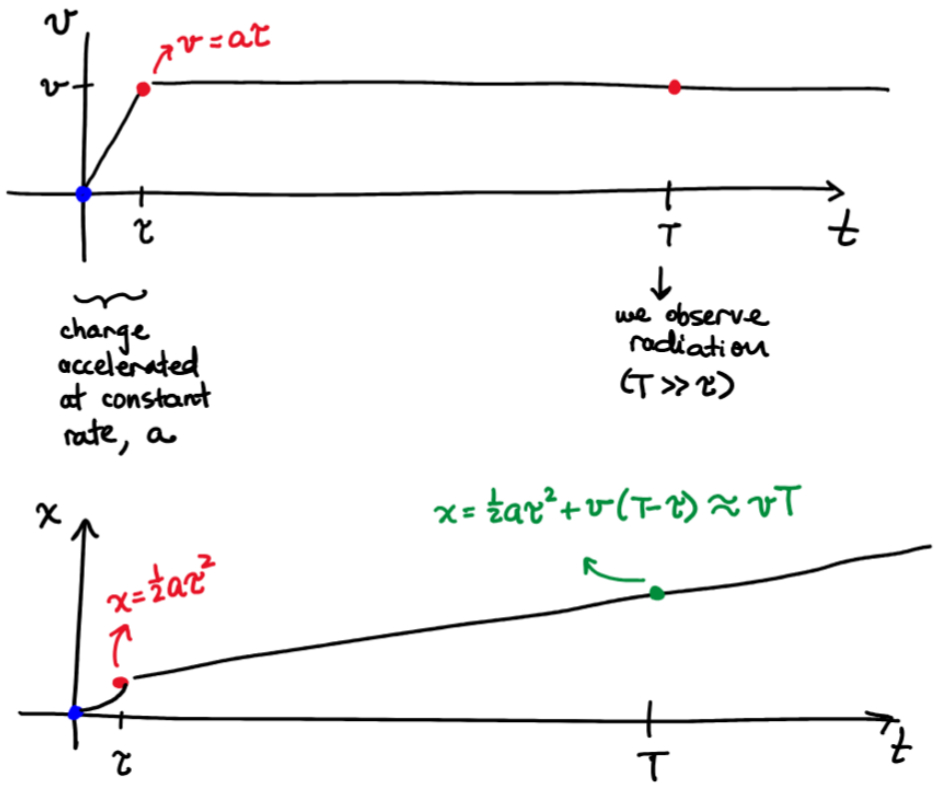
The Setup

1. A free charge is at rest at $x = 0$ for all time $t \leq 0$.

2. The charge is "kicked" and accelerated with a constant a for a short period of time τ in the $+x$ direction.

3. After the acceleration, the charge moves at a constant velocity $v = a\tau$ in the $+x$ direction forever ($t > \tau$).

4. We will observe the electric field lines far away from the charge and at a much later time T after the acceleration ($\tau \ll T$).



Relativistic Causality

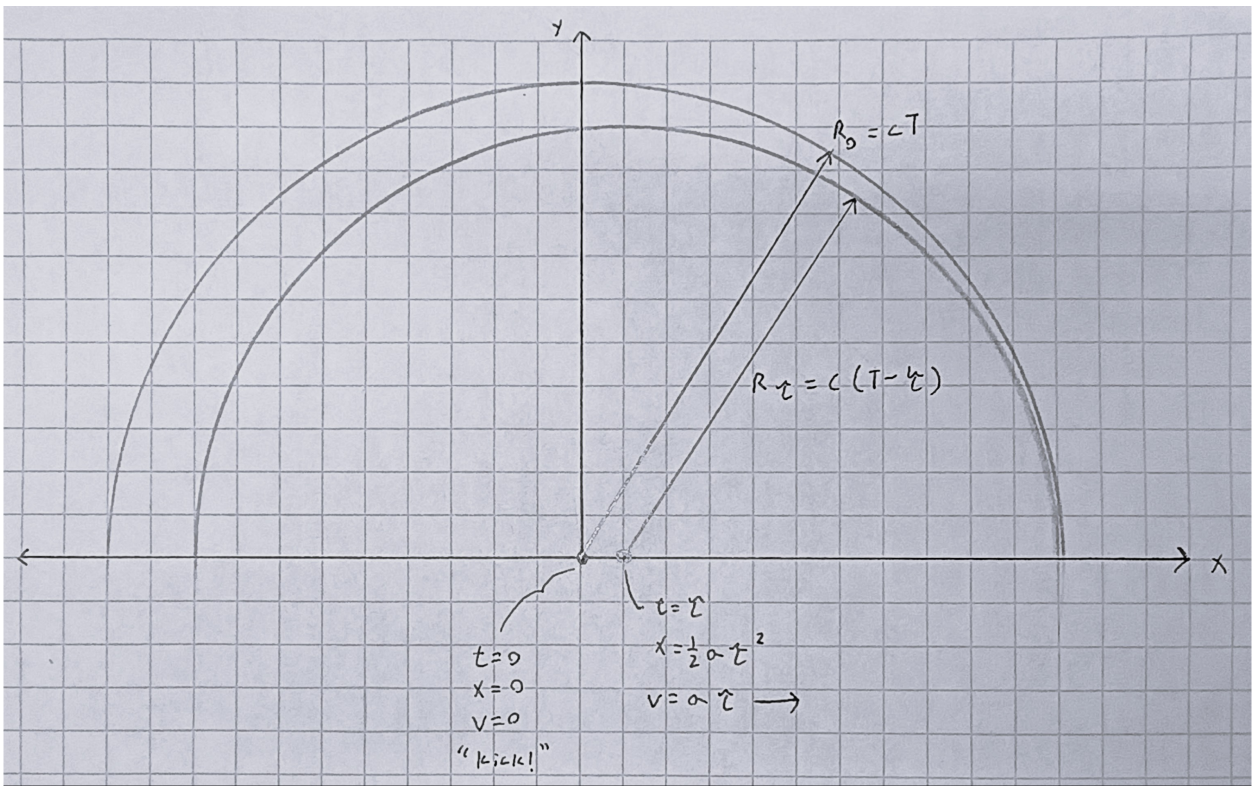
Relativistic causality states that no information can propagate faster than the speed of light. In this model, the information of interest is the beginning and end of the acceleration of the charge. These disturbances propagate outward at speed c.

The outer boundary, which represents the beginning of the acceleration, is a sphere of radius centered at the origin, corresponding to the physical position of the charge at $t = 0$.

$$R_0 = cT$$

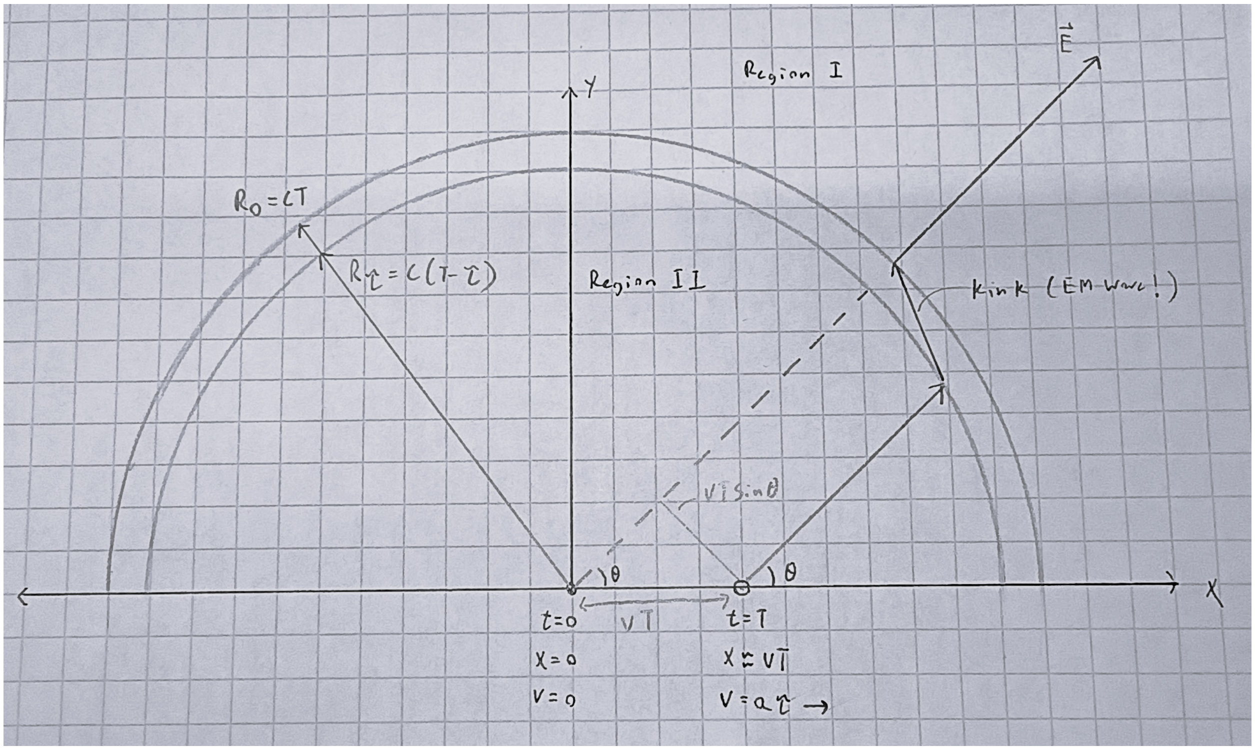
Likewise, the inner boundary, which represents the end of the acceleration, is a sphere of radius centered at $x = \frac{1}{2}a\tau^2$ which is the physical position of the charge at $t = \tau$.

$$R_1 = c(T - \tau)$$



The region between these two radii contains the kink in the electric field, that is, the electromagnetic wave. This is why we say that an accelerated charge produces radiation.

Under the approximation $\tau \ll T$ and in the limit $\tau \to 0$, the displacement of the charge during the acceleration becomes negligible. Thus, the charge's position after the acceleration is approximately the same as its initial position, and the two spheres may be treated as approximately concentric. Additionally, following our previous approximation that the electric field of a charge moving at constant velocity is essentially the same as that of a static charge, we can say that the field lines entering and exiting this region are parallel, forming the same angles to the x axis.



Due to relativistic causality, the electric field lines in region I must be that of a static charge at the origin (because the information of the acceleration has not yet reached this region). The electric field lines in region II, then, must be that of a charge moving at constant velocity (because the information of the acceleration has already reached this region).

The Radiated Electric Field

So what must the electric field be in the region in between, the spherical shell of thickness $c\tau$, which is the *only* region containing information about the charge *during its acceleration*? Since the electric field is continuous, the field lines must connect smoothly across this region. As a result, they develop a kink within the shell, giving the field both a radial component E_r and a transverse component E_θ.

We can derive the magnitudes of these components by using the geometry to decompose the electric field:

$$\vec{E}_a = E_r\hat{r} + E_\theta\hat{\theta}$$

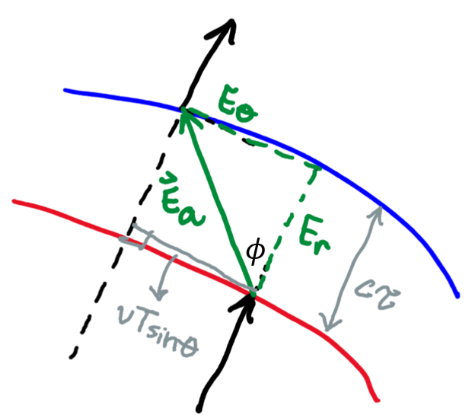
$$tan\phi = \frac{E_\theta}{E_r} \implies E_\theta = E_r\frac{vT\sin\theta}{c\tau}$$

Using Gauss's law, we can find E_r:

$$E_r = \frac{q}{4\pi\epsilon_0 R_0^2} = \frac{q}{4\pi\epsilon_0 (cT)^2}$$

Thus,

$$E_\theta = \frac{v}{\tau}\frac{q\sin\theta}{4\pi\varepsilon_0 c^3 T}$$



Note: We can choose any Gaussian surface that encloses the charge. Since the electric field line is continuous, its magnitude remains the same along its entire path. However, choosing a sphere that lies entirely within region I is most convenient, because the electric flux there comes solely from the static charge at the origin due to relativistic causality. In region II, by contrast, the charge has moved and is no longer centered within the sphere, leading to a non-uniform electric flux distribution.

A more useful form for E_θ can be obtained by substituting $a = v/\tau$ and $R = cT$:

$$E_\theta = \frac{1}{4\pi\varepsilon_0}\frac{qa\sin\theta}{c^2 R}$$

Therefore, the electric field kink is

$$\vec{E}_a = \frac{1}{4\pi\epsilon_0}\frac{q}{R^2}\hat{r} + \frac{1}{4\pi\varepsilon_0}\frac{qa\sin\theta}{c^2 R}\hat{\theta}$$

Since the radial component E_r falls off as $1/R^2$ while the transverse component E_θ falls off as $1/R$, for large observation distances R, $E_r \ll E_\theta$ and the electric field is dominated by the transverse component. Therefore, for radiation, and also for quantifying the resulting electromagnetic wave, we can use just E_θ, the transverse component of the electric field.

This also makes intuitive sense in our model. Since $R = cT$, a large R corresponds to a large T, so the approximation $\tau \ll T$ becomes increasingly valid. In this limit, the region between regions I and II effectively shrinks to a spherical shell of negligible thickness, and the field lines become squished so that they are essentially tangential to the sphere, meaning they are purely transverse.

Finally, putting all the intuition and math together, we have the formula for the **Thompson radiation** generated from an accelerating charge:

$$\boxed{\vec{E}(\vec{r},t) = \frac{1}{4\pi\varepsilon_0}\frac{qa\sin\theta}{c^2|\vec{r}|}\,\hat{\theta} \quad \text{(Thompson radiation)}}$$

Where:

- a is the acceleration of the charge.
- q is the charge of the particle.
- θ is the angle between the direction of acceleration and the direction of observation.
- c is the speed of light.

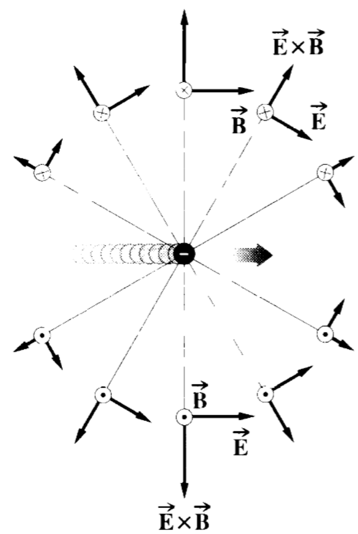
$\vec{r}$ is the position vector from the charge's initial position to the observation point, and is also the *direction of the electromagnetic wave propagation* $\vec{k}$. This is because the kink in the electric field lines spreads out radially outward, while the electric field itself is always tangential to $\hat{r}$.

Also, while time t doesn't explicitly appear in the formula physically, recall that the radius $|\vec{r}|$ is related to the time of observation T by $|\vec{r}| = cT$. Additionally, the acceleration a can also be a function of time, and the formula for Thompson radiation can be applied to each instantaneous value of a.

As a concluding thought, electromagnetic waves are generated by an accelerated charge fundamentally because of relativistic causality. The discrepancy between the travelling spheres of information about the beginning and the end of the acceleration simply can only be resolved by the electric field lines developing a kink in the region between the two spheres. From the arguments developed in the past four pages, this kink is not optional but required, and it is precisely this feature that constitutes the radiation field.

Note: Recall from the general plane wave solution that we can also define the magnetic field *induced* by the radiated electric field kink using

$$\vec{B}(\vec{r}, t) = \frac{1}{c}\hat{k} \times \vec{E}, \quad \hat{k} = \hat{r}$$



Poynting Flux

Using the Poynting flux vector, we can calculate the rate of energy radiated by the accelerating charge:

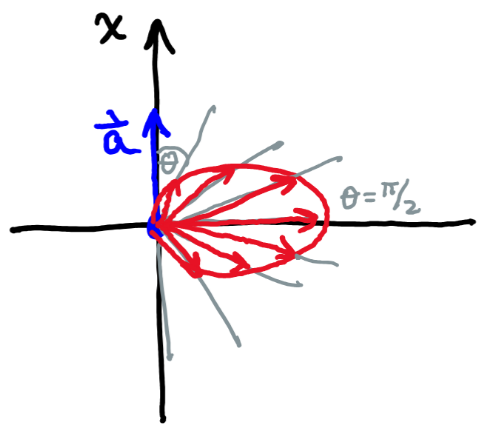
$$\vec{S}(\vec{r}, t) = \frac{1}{\mu_0}\vec{E} \times \vec{B}$$

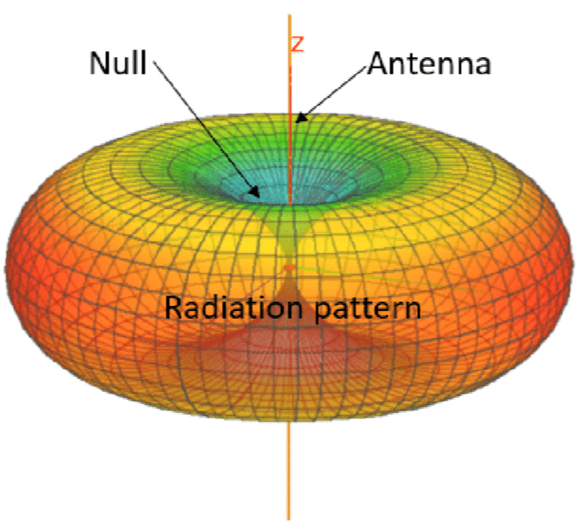
$$\boxed{|\vec{S}| = \varepsilon_0 c|\vec{E}|^2 = \frac{q^2 a^2 \sin^2\theta}{16\pi^2\varepsilon_0 r^2 c^3} \quad \left[\frac{\text{W}}{\text{m}^2}\right]}$$

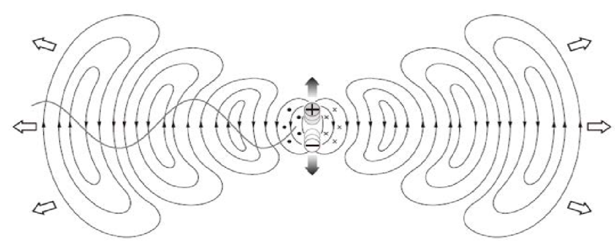
this is the power per unit area at a distance $r = |\vec{r}|$ from the charge and an angle θ from the direction of acceleration due to a short acceleration a.

What Does it Look Like?
The Poynting vector $\vec{S}$ points in the direction of electromagnetic wave propagation $\hat{r}$, so energy flows radially outward from the charge. Thus, the radiated power spreads away from the source in spherical wavefronts, like sound waves or ripples on a pond from an isotropic source.

Power depends on the angle θ, meaning that it is strongest at $\theta = \pi/2$ (perpendicular to the direction of acceleration) and weakest at $\theta = 0$ or π (along the direction of acceleration). This results in a doughnut (toroidal) radiation pattern, with the charge at the center and the hole of the doughnut along the axis of acceleration.



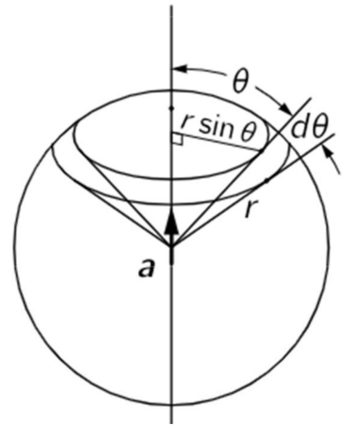




The $\vec{\mathbf{E}}$-field of an oscillating electric dipole.

Larmor Power

Larmor power is the total power radiated by an accelerating charge, which can be found by integrating the Poynting flux (power per area) over a spherical surface that contains the charge, thus capturing all the flux. The sphere is centered on the charge's initial position, and its radius r is arbitrary, since the total radiated power is independent of the distance from the charge.
Since $\vec{S}$ is points in the $\hat{r}$ direction and is always normal to the sphere surface,

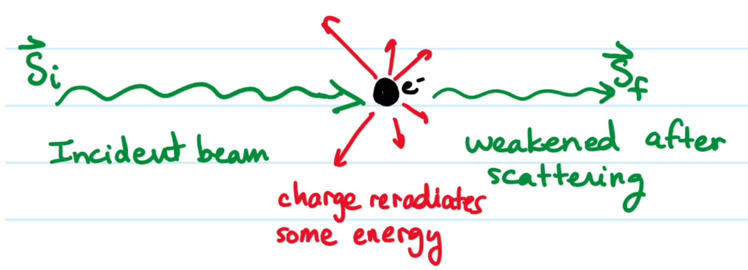


$$\begin{aligned}
P &= \oint_S \vec{S}\cdot d\vec{A} = \int |\vec{S}|dA \\
&= \int_0^{2\pi}\int_0^{\pi} |\vec{S}|r^2 \sin\theta d\theta d\phi \\
&= \int_0^{\pi} |\vec{S}|2\pi r^2 \sin\theta d\theta \\
&= \int_0^{\pi} \frac{q^2a^2\sin^2\theta}{16\pi^2\varepsilon_0 r^2c^3} 2\pi r^2 \sin\theta d\theta \\
&= \frac{q^2a^2}{8\pi\varepsilon_0 c^3} \underbrace{\int_0^{\pi} \sin^3\theta d\theta}_{4/3}
\end{aligned}$$

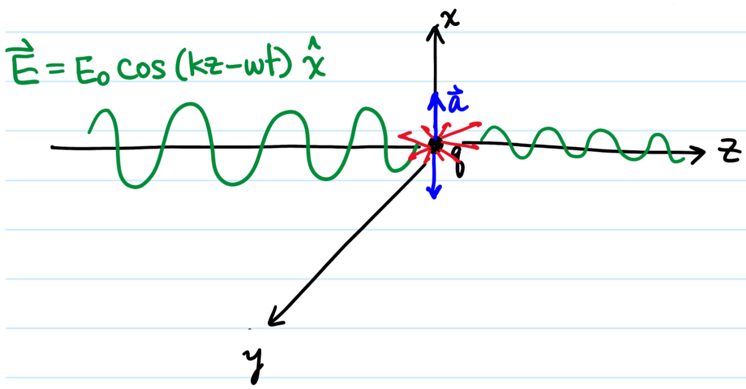
$$\boxed{P_{\text{rad}} = \frac{q^2a^2}{6\pi\varepsilon_0 c^3} \quad \text{(Watts)}}$$

Scattering by Free Charges

Scattering occurs when an incoming electromagnetic wave interacts with a free charge (Thompson scattering) or a bound charge (Rayleigh scattering). The incoming wave causes the charge to accelerate, which will then *reradiate* the energy in all directions. Our goal is then to quantify the scattered radiation in terms of its power and how the initial wave is changed by the scattering process.



Thompson Scattering
We will begin with Thompson scattering which describes the scattering of an electromagnetic wave by a free charge. Consider the following incoming electromagnetic wave with the electric field:



A free charge at the origin ($z = 0$) will experience a force equal to:

$$\vec{F} = q\vec{E} = qE_0 \cos(\omega t)\hat{x}$$

$$\implies a = a_x = \frac{qE_0}{m}\cos(\omega t)$$

Plugging this into the formula for Larmor power gives the instantaneous power radiated by the accelerating charge:

$$P(t) = \frac{q^2 a^2}{6\pi\varepsilon_0 c^3} = \frac{q^2}{6\pi\varepsilon_0 c^3}\left(\frac{qE_0}{m}\cos(\omega t)\right)^2 = \frac{q^4 E_0^2}{6\pi\varepsilon_0 m^2 c^3}\cos^2(\omega t) \quad \text{(Watts)}$$

Taking the time average gives the average power radiated by the accelerating charge (which is also the power of the scattered wave):

$$\boxed{\langle P_{\text{scatt}}(t)\rangle = \frac{q^4 E_0^2}{12\pi\varepsilon_0 m^2 c^3}}$$

Notice that $\langle P_{\text{scatt}}(t)\rangle$ is proportional to the square of the incident electric field amplitude E_0^2, thus we can rewrite the power of the scattered wave to be related to the intensity $I_{\text{inc}} = |\langle\vec{S}\rangle_{\text{inc}}|$ of the incident wave:

$$|\langle\vec{S}\rangle_{\text{inc}}| = \frac{1}{2}\varepsilon_0 E_0^2 c \qquad \implies \qquad E_0^2 = \frac{2|\langle\vec{S}\rangle_{\text{inc}}|}{\varepsilon_0 c}$$

$$\langle P_{\text{scatt}}(t)\rangle = \frac{q^4}{12\pi\varepsilon_0 m^2 c^3}\frac{2|\langle\vec{S}\rangle_{\text{inc}}|}{\varepsilon_0 c} = \frac{8\pi}{3}\underbrace{\left(\frac{q^2}{4\pi\varepsilon_0 m c^2}\right)^2}_{\text{area, m}^2}\underbrace{|\langle\vec{S}\rangle_{\text{inc}}|}_{\substack{\text{intensity}\\ \text{W/m}^2}}$$

Thompson Cross Section

We define the Thomson cross section σ_{Th} as the effective area that quantifies the likelihood of scattering by a free charge when an incident electromagnetic wave interacts with it. A useful mechanical analogy is to imagine shooting a stream of BB pellets at an 8 ball. If a BB hits the 8 ball, it bounces off, and the probability of this happening depends on the size of the 8 ball (a larger 8 ball increases the chance of being hit).

In this case, the Thomson cross section is not the *physical* cross-sectional area of the charge, since a charge doesn't exactly have a physical size. Instead, it is an *effective* area that captures the probability of scattering in the same intuitive way. Therefore,

$$\boxed{\langle P_{\text{scatt}}(t)\rangle = \sigma_{\text{Th}}|\langle\vec{S}\rangle_{\text{inc}}|}$$

Where

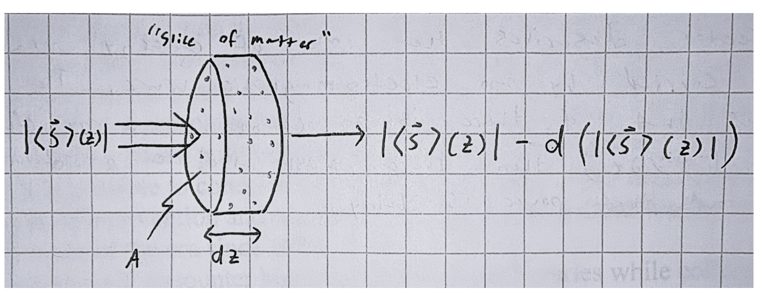
$$\boxed{\sigma_{\text{Th}} = \frac{8\pi}{3}\left(\frac{q^2}{4\pi\varepsilon_0 mc^2}\right)^2 \quad \text{(general cross-section)}}$$

$$\boxed{\sigma_{\text{Th}} = \frac{8\pi}{3}r_e^2 \quad \text{(electron cross-section)}}$$

Passing Through a Material

So far we have quantified the power of the scattered wave from *a single free charge.* To find the *total* power scattered by a volume that contains many free charges, we can simply multiply the power scattered per charge by the number of charges in the volume.

Consider a slice of matter with thickness dz, cross-sectional area A, and number density of free charges N (the number of free charges per unit volume), where each free charge has a Thomson cross section σ_{Th}. We aim to derive an expression for the change in intensity $d\Big(|\langle\vec{S}\rangle_{\text{inc}}|\Big)$ due to scattering by the free charges within this volume. This scattering removes energy from the forward direction, causing a decrease in the intensity of the transmitted wave.



The power entering the volume is given by: $|\langle\vec{S}\rangle(z)|\,A$

similarly, the exiting power from the volume is given by: $|\langle\vec{S}\rangle(z)|\,A - Ad(|\langle\vec{S}\rangle(z)|)$

Then we can define the total power lost by scattering in the volume as:

$$-A\,d(|\langle\vec{S}\rangle(z)|) = \underbrace{(NA\,dz)}_{\substack{\text{total number}\\ \text{of scatterers}}} \cdot \underbrace{\sigma_{\text{Th}}(\langle\vec{S}\rangle(z))}_{\substack{\text{power radiated}\\ \text{per scatterer}}}$$

We can solve this differential equation by separating variables and integrating:

$$\frac{d(|\langle\vec{S}\rangle(z)|)}{|\langle\vec{S}\rangle(z)|} = -N\sigma_{\text{Th}}dz \quad\implies\quad |\langle\vec{S}\rangle(z)| = |\langle\vec{S}\rangle(0)|e^{-N\sigma_{\text{Th}}z}$$

Since $|\langle\vec{S}\rangle(0)|$ is equal to the intensity of the incident wave $\langle\vec{S}\rangle_{\text{inc}}$, we can rewrite this as:

$$\boxed{|\langle\vec{S}\rangle(z)| = |\langle\vec{S}\rangle_{\text{inc}}|e^{-N\sigma_{\text{Th}}z} \quad \text{(decay of intensity due to scattering)}}$$

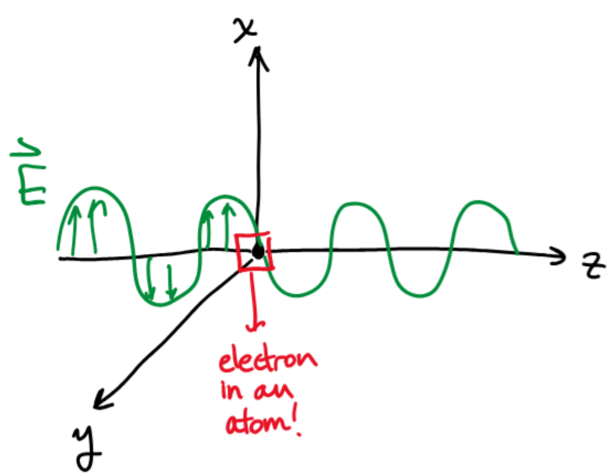
<u>Note:</u> This makes intuitive sense because an increase in the number of free charges, the thickness of the material, or the scattering cross section would all increase the likelihood of scattering, thus leading to a greater decrease in the intensity of the transmitted wave.

Scattering by Bound Charges

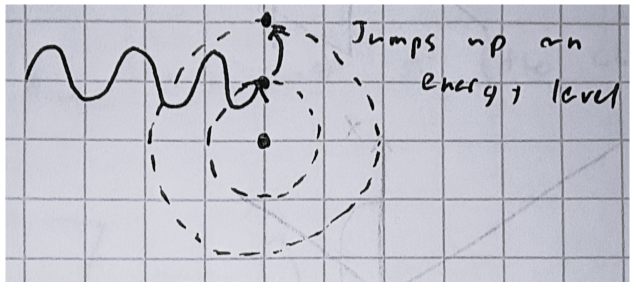
Similar to Thomson scattering, Rayleigh scattering describes the scattering of an electromagnetic wave by a *bound* charge. Instead of a free charge that moves freely in response to the incident wave, a bound charge experiences a restoring force due to its attraction with the nucleus. For our model, we will use Bohr's model of the atom, in which electrons occupy fixed circular orbitals around the nucleus, with radii determined by their energy levels.

Consider an incoming electromagnetic wave with the electric field:

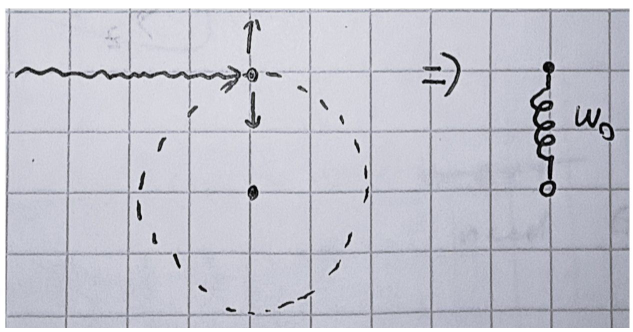
$$\vec{E} = E_0\cos(\omega t - kz)\hat{x} \implies \vec{E} = E_0\cos(\omega t)\hat{x} \quad \text{(at origin, z=0)}$$



Recall that in the Bohr model, if a valence electron receives enough energy from the incoming wave, it can be excited to a higher energy level or even ionized (escapes the atom completely).



However, if the incoming wave does not have enough energy to excite the electron to a higher energy level, the electron will simply oscillate in its current orbital in response to the electric field of the wave. For Rayleigh scattering, we are interested in this case, where the valence electron behaves like a driven harmonic oscillator with a natural frequency ω_0 based on the *characteristic frequency* of the atom.



Note: To simplify the model, we are only considering the acceleration of *one* electron in the atom, which follows the valence electron idea.

For a classical simple harmonic oscillator, the characteristic frequency is given as

$$\omega_0 = \sqrt{\frac{k}{m}}$$

We will use this characteristic frequency in our model instead of introducing a stiffness k, since it is more intuitive in the context of atomic behavior. Specifically, the **characteristic** or **transition** frequency ω_0 of an atom is the frequency of the electromagnetic wave that the atom emits or absorbs when it transitions between energy levels. It is often specified in terms of wavelength, but because it corresponds to an electromagnetic wave, we can express it as a frequency using the relation $c = \omega/k$.

Note: If the frequency of the incoming wave ω is close to the characteristic frequency ω_0, the electron undergoes resonance and may transition to a higher energy level, which is not the behavior we are modeling. As such in the derivation, the condition $\omega = \omega_0$ produces a singularity in the solution.

Solving for the Acceleration
The equation of motion for the driven harmonic oscillator is given by:

$$\sum F_x = m\frac{d^2x}{dt^2} = \underbrace{qE_0\cos(\omega t)}_{\vec{E}\text{ field}} - \underbrace{\omega_0^2 mx}_{\substack{\text{restoring}\\ \text{force}}}$$

$$\frac{d^2x}{dt^2} + \omega_0^2 x = \frac{qE_0}{m}\cos(\omega t)$$

Solving the differential equation for the steady-state particular solution gives the position of the electron as a function of time (guessing a solution of the form $x(t) = A\cos(\omega t)$, or using laplace transform with zero initial conditions):

$$x(t) = \frac{qE_0}{m(\omega_0^2 - \omega^2)}\cos(\omega t)$$

$$\implies a(t) = \frac{d^2x}{dt^2} = \frac{qE_0}{m}\left(\frac{\omega^2}{\omega^2 - \omega_0^2}\right)\cos(\omega t)$$

Larmor Power
Plugging this acceleration into the formula for Larmor power gives the instantaneous power radiated by the accelerating charge:

$$P(t) = \frac{q^2a^2}{6\pi\varepsilon_0 c^3} = \frac{q^4E_0^2}{6\pi\varepsilon_0 m^2c^3}\left(\frac{\omega^2}{\omega^2 - \omega_0^2}\right)^2\cos^2(\omega t) \quad \text{(Watts)}$$

$$\implies \langle P_{\text{scatt}}(t)\rangle = \underbrace{\frac{8\pi}{3}\left(\frac{q^2}{4\pi\varepsilon_0 mc^2}\right)^2}_{\sigma_{\text{Th}}}\left[\frac{\omega^2}{\omega^2 - \omega_0^2}\right]^2 |\langle\vec{S}\rangle_{\text{inc}}|$$

Where the *Rayleigh cross-section area* is simply a rescaling of the Thomson cross-section area by a factor

$$\sigma_{\text{Ray}} = \sigma_{\text{Th}}\left[\frac{\omega^2}{\omega^2 - \omega_0^2}\right]^2$$

$$\boxed{\langle P_{\text{scatt}}(t)\rangle = \sigma_{\text{Ray}}|\langle\vec{S}\rangle_{\text{inc}}|}$$